

ZIHAN LIM

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EXECUTIVE SUMMARY

- Quantitative Researcher with Agentic AI workflow integration, Machine Learning/Deep Learning/Reinforcement Learning expertise, Sovereign Wealth Fund experience, delivering a 35.8% annualized return via a regime-switching portfolio model. Skilled in Python, C++, and time series analysis, with proven success in algorithmic trading and data-driven strategies for global markets. Agile problem-solver with experience in high-performance teams across Singapore, Beijing, and Hong Kong.

KEY SKILLS

AI Agent Orchestration (LangGraph, OpenClaw) | Python | C++ | R | Algorithmic Trading | Risk Management
Machine Learning | Deep Learning | Financial Time Series | Financial Modelling | SQL | Econometrics | Java

WORK EXPERIENCE

Binance | Data Scientist & AI Engineer | Singapore, SG

JUL 2026 – CURRENT

- Market Manipulation Risk Engine—Binance Futures and Spot (python, Sagemaker, SQL)**
 - Formulated price-manipulation detection as a two-sided estimation problem:** ex-ante feasibility (cost-to-move functional over order-book/mark microstructure) and ex-post now-casting (coordinated-attack detection from the trade tape); deployed as a single SageMaker BYOC endpoint across spot and perpetual futures.
 - Built a venue-correct feasibility model:** walk-the-book buy-cost for spot vs. an index-anchored mark-movability model for perpetuals — a perp is only manipulable if its mark can be dislodged from the index — output as interpretable cost-to-move in USD to move price 1%.
 - Designed a validation harness most detection pipelines lack:** Market-cap-matched control group + majors negative-control, achieving 96% detection recall at 4.2% false-positive rate with structurally un-manipulable majors correctly excluded.
 - Identified pre-attack prediction as fundamentally gated on account-coordination data (same-entity/crew detection)

GIC Singapore Sovereign Wealth Fund | Quantitative Researcher & AI Engineer Off-cycle | Singapore, SG

FEB 2026 – JUL 2026

- Kartharsis:** IM/VM risk co-pilot with UI/UX and quantitative model proposal for CCP margin forecasting; later validated when external vendors (OpenGamma included) moved in the same direction. Developed the RC-FHS-QR framework: a 5-layer architecture for Initial Margin forecasting under market stress, validated with 5-fold rolling-window cross-validation against CME IM data.
 - Layer 1:** Hybrid HMM + Random Forest ensemble for market regime classification (Calm/Volatile/Stressed)
 - Layer 2:** Filtered Historical Simulation + Quantile Regression + Extreme Value Theory for IM magnitude
 - Layer 3:** Competing Risks Survival Analysis (Kaplan-Meier, Cox PH, Fine-Gray) for IM timing/lag prediction
 - Layer 4:** GARCH-t with Student-t innovations for Variation Margin forecasting
 - Layer 5:** t-Copula + Quasi-Monte Carlo (Sobol sequences) for joint tail dependence and LaR fan charts
- IM-Jarvis:** Agentic investment-management platform built by directing AI coding agents through structured workflows with precise requirements, documented architectural decisions, and numerical-parity verification.
 - Five domain agents (compliance, risk, portfolio, reporting, trade) on TimescaleDB/PostgreSQL + FastAPI + React; every portfolio mutation in an immutable audit trail; agents can research and propose but not execute — deterministic validation gate + human approval required for every order. Presented roadmap to Head of Enterprise Risk Management for approval by Chief Risk Officer.
- Digital Agora** a full-stack application embedding a curated roster of philosophers and financiers grounded in their documented corpus, running a multi-provider AI architecture with a documented retrieval-augmented generation pipeline; the platform served as both a dialogue engine and an analytical tool for structured forecasting.
- Research + infrastructure:** Produced a fully-cited research study on post-GFC short-term borrowing halts and a technical brief on India's 2026 capital-control package, delivered as reports and slide decks. Built and maintained a live Federal Reserve Bank of New York data pipeline in support of the team's T-bill repo playbook. All outputs passed through human-in-the-loop review and a critique-and-refine loop before sign-off.

FOSTER + PARTNERS | Architectural Designer (Architect) | Singapore, SG

AUG 2022 – JUN 2024

- Developed concept design and construction documents for a proposed 60,000 sqm Headquarters for Philippines' largest telecom company PLDT, housing 5000 employees.
- Designed an eco-friendly high-rise luxury condominium in Bangkok with high sustainability. Reduced embedded carbon by 46%; received client commendation for efficient design, maximizing returns on investment.
- Utilized Computational design tools and Building Information Models to streamline workflow, achieving design goals while increasing overall efficiency by 30%.

UNSTUDIO LIMITED | Junior Architectural Designer (Junior Architect) | Hong Kong, HK

MAY 2021 – AUG 2022

- Designed and drafted a proposed 200,000 sqm Tencent Shenzhen Headquarters for 12,000 employees to large-scale 900 units condominium in Kuala Lumpur and executed on all phases of design and construction process on timeline and within budget with client satisfaction rate of over 95%.
- Collaborated with technical specialists, completing detailed design; Designed, modelled, and performed façade optimization as means for value engineering, reducing construction cost by 34% and improving a building's energy efficiency by 16%

MAD ARCHITECTS | Architectural Assistant | Beijing, BJ

AUG 2019 – AUG 2020

- Developed and executed 11 key design projects, such as Shenzhen Bay Cultural Plaza, Hainan Science and Technology Museum, Jiaxing Civic Centre and others from schools to museums, cultural centres, and hospitals.

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PROJECTS

DEEPCANDLENET: VISION TRANSFORMER AND PATCHTST-DRIVEN CANDLESTICK PATTERN TRADING STRATEGY	APR 2025 – AUG 2025
- Developed a Python-based trading system integrating Vision Transformer (ViT) and PatchTST to detect candlestick patterns and forecast trends in multi-asset financial data (stocks, cryptocurrencies) from yfinance and CCXT. - Engineered features such as OHLC, shadow lengths, and GAF images, leveraging ViT for pattern recognition and PatchTST for multivariate time series prediction, with patterns labeled using TA-Lib and Dynamic Time Warping. - Optimized trading signal accuracy using transfer learning for ViT and Adadelata optimization for PatchTST, enhancing performance over traditional models on a 2017–2023 dataset. - Implemented a backtesting framework with risk controls (stop-loss, position sizing) and visualized candlestick data with matplotlib, incorporating explainable AI to refine pattern classification and target profitability.	
REGIME-BASED SMART BETA PORTFOLIO OPTIMIZATION WITH FACTOR INVESTING	FEB 2025 – MAR 2025
Achieved 35.8% annualized return (Sharpe 1.67) by designing a Python-based Smart Beta portfolio for MAANG stocks, integrating regime-switching models with multi-factor data (2015–2019).	
- Built a Python-based backtesting framework for a Smart Beta portfolio of MAANG stocks, leveraging pandas and NumPy to process and analyze multi-factor data (Momentum, Value, Quality, LowVol) from 2015–2019. - Designed a regime-switching model integrating market signals (VIX, S&P 500) and macro data (Treasury yields), achieving 35.8% annualized return with a 1.67 Sharpe ratio. - Optimized factor weights using SciPy's SLSQP, applying constraints to balance risk and return, and visualized portfolio dynamics with matplotlib for monthly rebalancing insights. - Debugged and refined code to resolve performance discrepancies, demonstrating strong problem-solving and analytical skills.	
DYNAMIC MACD TRADING STRATEGY WITH KELLY OPTIMIZATION	JAN 2025 – FEB 2025
Developed a Kelly-optimized trading system in Python, yielding 28.9% annualized returns (Sharpe 2.36) for MAANG stocks, with real-time signal plotting and risk controls.	
- Engineered a Python-based trading system for MAANG stocks, processing historical data (2015–2019) with pandas, yfinance, and ta to compute MACD, EMA, and ATR indicators. - Designed a Kelly Criterion-based position sizing model using NumPy, dynamically adjusting weights across 5 stocks to maximize returns (28.9% annualized) while capping risk. - Analyzed performance with metrics (Sharpe: 2.36, max drawdown: -9.64%) and visualized portfolio value, drawdowns, and stock weights using matplotlib. - Optimized strategy logic to prevent leverage, integrating real-time signal plotting and benchmark comparisons (S&P 500).	

EDUCATION

Master of Science in Digital Financial Technology NATIONAL UNIVERSITY OF SINGAPORE	AUG 2023 – DEC 2025
Relevant Coursework: Machine Learning for Finance, Object Oriented Programming, Data Structures and Algorithms, Software Development, Fintech Innovations for Consumers, Digital Transformation at Financial Institutions, Financial Modelling, Algorithmic Trading Systems Design and Deployment, Deep Learning for Finance, Electronic Financial Markets, C++ in Financial Engineering, Analytics for Capital Market Trading	
Master of Science in Financial Engineering (Part-time) WORLDQUANT UNIVERSITY	JUN 2024 – JUN 2026
Relevant Coursework: Financial Markets, Financial Data, Financial Econometrics, Derivative Pricing, Stochastic Modeling, Machine Learning in Finance, Deep Learning in Finance, Portfolio Management, Risk Management	
Advanced Professional Certificate in Software Engineering NANYANG TECHNOLOGICAL UNIVERSITY	JUN 2023 – NOV 2023
Relevant Coursework: Object Oriented Programming, Java, Spring Framework, JavaScript, React.js, HTML5, CSS, React Native, CICD, Database Management	
Master of Architecture & Bachelor of Arts in Architecture (HONOURS) (GPA 4.59/5) NATIONAL UNIVERSITY OF SINGAPORE	AUG 2014 – MAY 2019
Relevant Coursework: Parametric Design: Algorithmic Logic, Python Scripting, Rule-based Modeling. Structural Systems: Applied Physics, Static Equilibrium, Vector Analysis. Environmental Modeling: Data Simulation, Stochastic Modeling, Numerical Analysis. BIM Management: Data Architecture, Large-scale Database Management. Professional Practice: Project Risk Management, Resource Allocation, Budget Optimization.	

AWARDS AND HONOURS

NUS Fintech Society – Quantitative Researcher	2024
- Developed Automated Martingale Forex Trading Strategy	
NUS Blockchain Pitch Night – Asian Institute of Digital Finance Winner	2023
NUS Venture Initiation Programme Grant \$10,000 – NUS Enterprise	2023
- Co-founded Fury web3 wallet start-up in university with group of 3 Engineers and Designer; devised strategies for product development, building prototypes and develop business model.	
Solana Hyperdrive Blockchain Hackathon	2023
Selected NUS thesis representative at National Cheng Kung University (NCKU), Taiwan	2019
NUS-JTC i3 Centre Innovation Medal and Prize for Architecture (Best Design)	2018
- Winner of National First Prize in developing best design project addressing the theme of innovation in the design of industrial facilities and infrastructure, focusing on sustainability through Modular Integrated Construction (MiC).	